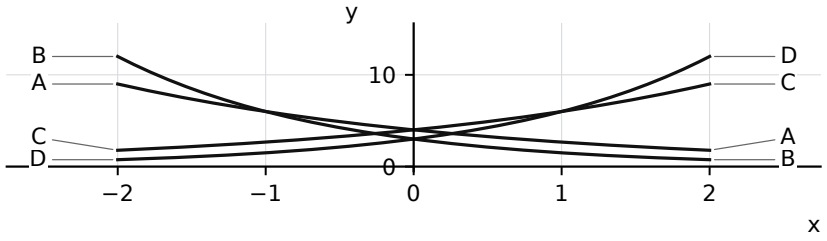
MATH 101 Final Exam

Practice 2 | Version 202610122020.47

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

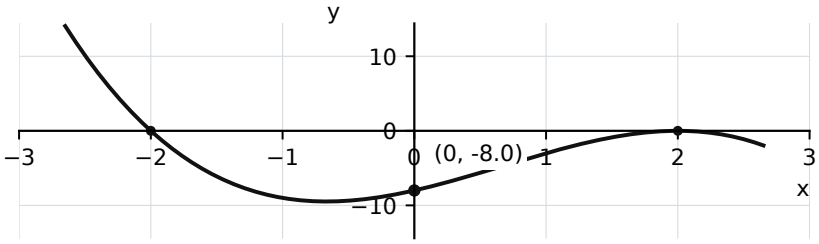
QUESTIONS	ANSWERS
1. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts. $f(x) = \frac{(x+3)(x+4)}{(x-6)(x+3)}$	1. (a) Domain 2 pts (b) Hole 2 pts (b) Vertical 1 pts (b) Horizontal 1 pts (c) x-intercept 2 pts (c) y-intercept 2 pts
2. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 4(\frac{3}{2})^x$ $(b)\quad 4(\frac{2}{3})^x$ $(c)\quad 3(2)^x$ $(d)\quad 3(\frac{1}{2})^x$ 	2. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts
3. Which pair satisfies h(x) = f(g(x))? $h(x) = (4x+4)^2 + 5$ $A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 4$ $B.\quad f(x) = 4x + 4,\ g(x) = x^2 + 5$ $C.\quad f(x) = x^2 + 4,\ g(x) = 4x + 5$ $D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 4$	3. Choice 10 pts

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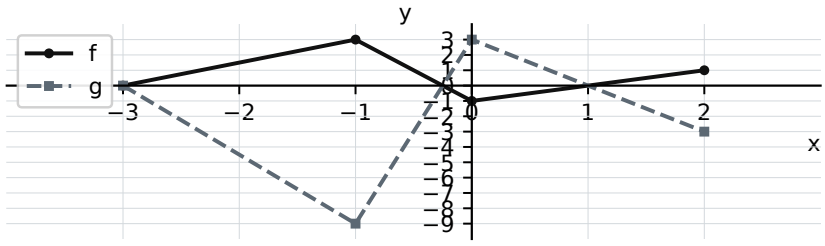
QUESTIONS	ANSWERS
4. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	4. Polynomial 10 pts
5. Find all zeros: (a) list Rational Root Theorem candidates, then test only -2 and 2; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros. $p(x) = x^3 + 5x^2 + 7x + 2$	5. (a) Candidates/root 2 pts (b) Quotient 4 pts (c) Other zeros 4 pts
6. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. $f(x) = 2\sqrt{x+9} - 6$	6. (a) Domain 2 pts (a) Range 2 pts (b) x-intercept 1 pts (b) y-intercept 1 pts (c) Infinite end 3 pts (c) Finite end 1 pts
7. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10. $(a)\quad \log(0.8) + \log(1250)$ $(b)\quad \frac{\log_7(64)}{\log_7(16)}$	7. (a) 5 pts (b) 5 pts

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QUESTIONS	ANSWERS
8. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain. $f(x) = \log_3(x-3) + 2$	8. (a) Domain 2 pts (a) Range 2 pts (b) Asymptote 1 pts (b) x-intercept 1 pts (c) Left end 2 pts (c) Right end 2 pts
9. Solve the nonlinear system algebraically. $y = x^2$ $y = x + 2$	9. Intersections 10 pts
10. The solid graph is f and the dashed graph is g. Select the expression for g(x). *A*. $g(x) = 3f(x)$ *B*. $g(x) = f(-3x)$ *C*. $g(x) = -3f(x)$ *D*. $g(x) = f(x) - 3$ 	10. Choice 10 pts

Instructor answer key: MATH 101 Final Exam

Practice 2 | Version 202610122020.47

Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a) Domain
$\mathbb{R} \setminus \{-3, 6\}$ — 2 pts

(b) Hole
$(-3,\ -\frac{1}{9})$ — 2 pts

(b) Vertical
$x = 6$ — 1 pts

(b) Horizontal
$y = 1$ — 1 pts

(c) x-intercept
$(-4,\ 0)$ — 2 pts

(c) y-intercept
$(0,\ -\frac{2}{3})$ — 2 pts

2. (a)
C — 3 pts

(b)
A — 2 pts

(c)
D — 3 pts

(d)
B — 2 pts

3. Choice
A — 10 pts

TEST PAGE 2

4. Polynomial
$-1(x+2)(x-2)^2$ — 10 pts

5. (a) Candidates/root
$\{-2, -1, 1, 2\};\ x = -2$ — 2 pts

(b) Quotient
$x^2 + 3x + 1;\ R = 0$ — 4 pts

(c) Other zeros
$x = -\frac{3}{2} - \frac{\sqrt{5}}{2},\ -\frac{3}{2} + \frac{\sqrt{5}}{2}$ — 4 pts

6. (a) Domain
$[-9, \infty)$ — 2 pts

(a) Range
$[-6, \infty)$ — 2 pts

(b) x-intercept
$(0,\ 0)$ — 1 pts

(b) y-intercept
$(0,\ 0)$ — 1 pts

(c) Infinite end
$x \to \infty :\ f(x) \to \infty$ — 3 pts

(c) Finite end
$x \to -9^{+} :\ f(x) \to -6$ — 1 pts

7. (a)
3 — 5 pts

(b)
$\frac{3}{2}$ — 5 pts

TEST PAGE 3

8. (a) Domain
$(3, \infty)$ — 2 pts

(a) Range
$(-\infty, \infty)$ — 2 pts

(b) Asymptote
$x = 3$ — 1 pts

(b) x-intercept
$(\frac{28}{9},\ 0)$ — 1 pts

(c) Left end
$x \to 3^{+} :\ f(x) \to -\infty$ — 2 pts

(c) Right end
$x \to \infty :\ f(x) \to \infty$ — 2 pts

9. Intersections
$(-1,\ 1);\ (2,\ 4)$ — 10 pts

10. Choice
C — 10 pts